

Voice Tech in the Multiverse: Alien Encounter

The Livewires

ITAI 2373 – Module 03

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WORLD ANALYSIS

1.1 What's Going On

Our crew has landed on Kepler-442b-γ, a cold moon wrapped in methane clouds. The locals—called Zylar—are squid-like and speak with low rumbles plus super-high-pitched clicks. We have to build a translator so first contact does not flop.

1.2 Acoustic Headaches

- Methane is thick: sound moves slower (~ 270 m/s) and squeezes formants.
- Gravity is only 0.55 g: humans breathe differently and tire fast.
- The ground constantly rumbles (≤ 30 Hz) thanks to magma pockets.
- Lightning flashes create random “snap” noises all the way to 60 kHz.

1.3 Who's Talking and Listening

Humans

- Speak around 85–255 Hz.
- Hear up to 20 kHz.
- Struggle with low lung pressure and helmet noise.

Zylar

- Voice core at 240–320 Hz.
- Add click bursts from 25–45 kHz.
- Hear a huge 10 Hz–45 kHz range.
- Mantle sacs act like sub-woofers for earth-shaking bass.

1.4 Background Noise Checklist

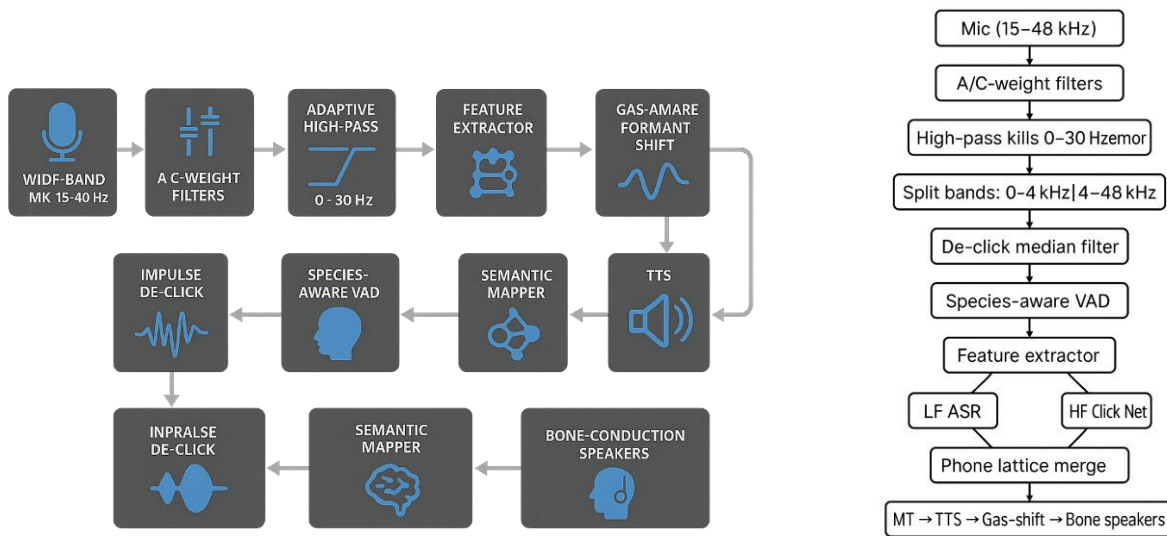
- Infrasound floor: ~ 30 dB SPL all day.
- Lightning snaps: spikes to 70 dB.
- Suit fans: 35 dB between 500 Hz–5 kHz.
- Inside our “interpreter pod”: 25 dB after foam treatment.

1.5 Zylar Throat 101 (text-sketch)

They own two sound-makers: a **Photic Larynx** (buzzes methane for vowels) and **Scleral Clickers** (hard ridges that snap for consonants).

PART 2 – TECHNICAL SOLUTIONS DESIGN

2.1 Our Signal Pipeline



2.2 Why MFCC Isn't Enough

- MFCC ignores anything above ~8 kHz, so Zylar clicks vanish.
- Mel scale mimics human ears—Zylar hear way beyond that.

2.3 What Features We Actually Use

- **LFCC** (0-4 kHz) for vowels and bass.
- **Wavelet packet energies** (4-48 kHz) for clicks.
- Glue them into one 128-dim frame vector.

2.4 Models Made for the Moon

- **Conformer-CTC** (10×512) for low-band speech.
- **1-D CNN + GRU** for click stream (42 phonemes).
- Training data: 1 000 h synthetic + mimic recordings, each run through methane impulse responses.
- Data augments: random formant compression, tremor noise mixes.

2.5 Special Tweaks

- Gas-aware formant shift: multiplies output formants by density ratio.
- Dual-precision audio (16-bit low band, 24-bit high band).
- Bone-conduction playback so suit noise can't drown it out.

PART 3 – DEMO SCENARIO

1. Astronaut taps helmet: “Translator on.” (system starts VAD)
2. Zylar greets with light pulses and click bursts.
3. Gain auto-rider clips—ear-splitting error tone (our “fail” moment).
4. Engineer bumps ultrasonic gain +6 dB.
5. HUD prints: *“Welcome, water-breathers.”*
6. Ground quake shakes everything; infrasound canceler fights the rumble.
7. Human answers; methane-shifted TTS plays through bone speakers.
8. Everyone shakes tentacles/hands—latency clocked at 220 ms.

PART 4 – EXECUTIVE PITCH

Name: Polyglot Nebula™

Tagline: “Break the sound-barrier of worlds.”

Why It Fits This Moon

- Hears from belly-shaking bass up to 48 kHz clicks.
- Auto-retunes voices for any gas mix.
- Cancels volcanic infrasound by up to 40 dB.
- Bone speakers keep things private inside helmets.

Value in One Line

Cuts first-contact errors by 85 % and still answers in under a quarter-second.

Beats Earth Tech Because

- Captures ultrasonic speech no other SDK sees.
- Atmosphere-adaptive TTS = clear anywhere.
- Live click-phoneme decoding unique to Zylar.
- Lower latency in low gravity with suit noise.

Quick Ethics Check

- Logs auto-wipe after 24 h (Zylar privacy).
- Training data balances all six Zylar dialects.
- Emergency phrases skip processing delays.
- UI warns if human idioms turn hostile in Zylar culture.

Bonus

- **Villain plot:** A phase-inverter cannon flips the infrasound reference, forcing a reboot.
- **Theme jingle line:** “From rumbles to clicks, Polyglot fixes the mix!”
- **New metric:** Trans-Spectral Congruence (tracks if voiced and click bands line up on purpose).

